

VENKATA MADHAV RAO VICHARAPU
SNOWFLAKE DEVELOPER

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Professional Summary:

- **Experienced Snowflake Developer** with extensive experience in designing and implementing scalable data models and ETL/ELT pipelines leveraging Snowflake, ensuring high-performance data storage, processing, and real-time data ingestion using Snowpipe, Streams, and Tasks.
- Expertise in **data integration** across multiple platforms, utilizing tools such as Snowflake SQL, dbt, Informatica, and cloud services like AWS (S3, Lambda, Glue, Redshift) and Azure (Data Factory, Blob Storage, SQL Database).
- Proficient in creating and optimizing **high-performance data models** (Star, Snowflake schemas) within Snowflake, and skilled in query optimization, clustering, materialized views, and resource management to enhance performance and reduce costs.
- Strong understanding of **data security and governance**, with experience in implementing role-based access control (RBAC), data encryption (at rest and in transit), and secure data sharing within Snowflake, ensuring strict compliance with privacy and security regulations.
- **Skilled in cloud infrastructure management** and automation using AWS, Azure, and Snowflake, including the setup of serverless ETL pipelines with AWS Lambda and Azure Data Factory, streamlining data workflows and reducing manual intervention.
- Adept in **data visualization** and business intelligence, developing interactive dashboards and reports using Power BI and Tableau, while integrating with Snowflake to provide real-time insights for business decision-making.
- Proven ability to **optimize performance** of large datasets and complex data workflows, using techniques such as partitioning, clustering, and efficient SQL querying, along with monitoring and managing system costs and resource usage.
- Experience working with both **structured and semi-structured data**, using Snowflake's variant data types to efficiently manage and query JSON, XML, and AVRO formats.
- Demonstrated ability to collaborate with **cross-functional teams**, ensuring the seamless flow of data from source to target systems while meeting business requirements and ensuring data integrity and quality through automated checks and validation.

Technical Skills:

Category	Technologies/Tools
Data Warehousing	Snowflake, Snowflake SQL, Snowpipe, Streams, Tasks, Virtual Warehouses, Materialized Views, Secure Data Sharing, Clustering Keys, Partitioning, External Tables, Snowflake Data Models (Star, Snowflake Schema)
ETL/ELT Tools	Informatica PowerCenter, AWS Glue, Azure Data Factory (ADF), dbt, Power BI, Tableau, Informatica Data Quality (IDQ), AWS Lambda, AWS Redshift, Control-M
Cloud Platforms	Snowflake, AWS, Azure
Programming Languages	SQL, Python, PL/SQL, Shell Scripting, JavaScript
Database/Big Data	Oracle, SQL Server, MySQL, PostgreSQL, Azure SQL Database, AWS Redshift, Snowflake
Data Integration	Kafka, Spark, Informatica, Python, Snowflake Connectors, dbt
Data Visualization/BI	Power BI, Tableau

Version Control/CI/CD	Git, dbt, CI/CD Pipelines
Security & Governance	Role-Based Access Control (RBAC), Data Encryption (at rest and in transit), Snowflake Security Features, Azure Active Directory (SSO), IAM, Data Quality Checks, Secure Views
Cloud Infrastructure	Azure (Azure Storage, Azure Data Factory, Azure Event Hubs), AWS (AWS S3, AWS Lambda, AWS Glue, AWS CloudWatch, IAM, Redshift)
Performance Optimization	Snowflake Query Profiling, Query Tuning, Materialized Views, Snowflake Clustering, Snowflake Query History, Snowflake Resource Monitoring, Azure Monitor, Cost Management Strategies
API/Integration	RESTful APIs, Snowflake Connectors (Python, Spark, Kafka), Azure API Integration
Automation	Snowpipe, AWS Lambda, Snowflake Tasks, dbt, Azure Data Factory (ADF), Control-M, Cron Jobs, CloudWatch

Professional Experience:

Client: T-Mobiles

Aug 2024 – Till Date

Role: Sr. Snowflake Developer

Responsibilities:

- Led the design and implementation of scalable, high-performance data models within Snowflake, using techniques such as **Snowflake's Snowpipe, streams, and tasks** for near real-time data ingestion and automated transformation. Utilized **Snowflake's variant data type** for storing semi-structured data (JSON, XML, AVRO) efficiently.
- Built and optimized **ETL/ELT pipelines** using **Snowflake SQL** and **Snowflake Task Scheduling** to orchestrate complex workflows. Integrated Snowflake with external tools like **Informatica** and **dbt** for automated data processing and transformation tasks.
- Designed advanced **SQL queries** using **Snowflake's window functions, CTEs, and materialized views** to transform large datasets efficiently. Leveraged **Snowflake's clustering and data pruning techniques** for performance optimization, ensuring minimal resource consumption and faster query execution.
- Optimized query performance by fine-tuning **virtual warehouses**, utilizing **query profiling** tools in Snowflake, and implementing **result caching**. Applied best practices in **clustering keys** and **automatic clustering** to further reduce query time and increase data access speeds.
- Managed **role-based access control (RBAC)** for Snowflake users, ensuring secure data access. Implemented **data encryption** both at rest and in transit using Snowflake's native security features. Created secure data sharing environments through **Snowflake Data Sharing** to seamlessly collaborate with partners while maintaining strict access control policies.
- Integrated Snowflake with third-party applications and platforms using **Snowflake connectors** (e.g., **Python, Spark, Kafka**). Built and maintained **RESTful APIs** to expose Snowflake data for consumption by internal and external systems.
- Enabled secure and scalable data sharing across departments and external organizations by utilizing **Snowflake Secure Data Sharing**. Created **secure views** and data access policies to ensure that only authorized users could access sensitive data.
- Automated data load processes using **Snowpipe** for real-time data ingestion and **Snowflake tasks** for scheduling recurring transformations and maintenance jobs, minimizing manual intervention and streamlining operations.
- Developed and implemented **data quality checks** using Snowflake's **SQL** capabilities to identify and rectify data issues during loading and transformation phases. Leveraged **dbt** for version-controlled transformation pipelines and continuous integration/continuous delivery (CI/CD) practices.
- Monitored Snowflake credit consumption and storage costs using **Snowflake's Resource Monitors and Query Profiling** tools. Reduced costs by selecting optimal virtual warehouse sizes, auto-scaling features, and implementing **query optimizations** that ensure high performance with minimal resource usage.

Tools Utilized: Snowflake, Snowpipe, Streams, Tasks, Variant Data Types, Snowflake SQL, dbt, Informatica, Power BI, Tableau, Kafka, Python, Spark, Resource Monitors, Query Profiling, Role-Based Access Control (RBAC).

Client: Credit Suisse

April 2023 – July 2024

Role: Snowflake Developer

Responsibilities:

- Designed and developed scalable, high-performance data models using **Snowflake** (Star, Snowflake schemas).
- Wrote efficient **Snowflake SQL** queries, creating **views**, **materialized views**, and **stored procedures** for various data processing needs.
- Implemented data transformation processes with **SQL** and **Snowflake Streams & Tasks** for real-time data processing.
- Integrated data from multiple sources, including **Azure Blob Storage** and **Azure Data Lake** into Snowflake using **Snowpipe** for automated data ingestion.
- Developed **ETL/ELT workflows** using **Azure Data Factory (ADF)**, automating data movement from on-prem databases and cloud sources into **Snowflake**.
- Utilized **Azure Data Lake** and **Azure Blob Storage** as data landing zones, leveraging **Snowflake's native connectors** for seamless integration.
- Implemented batch and real-time data processing using **Snowflake** and **Azure Data Factory** integration with **Azure Event Hubs**.
- Optimized **Snowflake query performance** by leveraging clustering keys, partitioning strategies, and proper data model design.
- Tuned **Snowflake Virtual Warehouses** for optimal scaling based on workload requirements (e.g., scaling up/down, auto-suspend, and auto-resume).
- Implemented best practices in **query optimization** by analyzing query plans and leveraging **Snowflake Query Profile** for troubleshooting.
- Configured **Azure Cloud** infrastructure for Snowflake integration, using **Azure Storage** and **Azure SQL Database** as staging areas for data ingestion.
- Integrated **Azure Data Factory** with **Snowflake** to create end-to-end automated workflows for data pipelines.
- Managed and monitored Snowflake performance and cost by leveraging **Azure Monitor** and **Snowflake's Resource Monitoring**.
- Implemented **role-based access control (RBAC)** and **Snowflake Security** features to ensure data access security, including **data encryption** in transit and at rest.
- Applied best practices for data governance and compliance by using **Snowflake Secure Data Sharing**, secure views, and access policies.
- Worked closely with **Azure Active Directory** for single sign-on (SSO) and user management across the Snowflake environment.
- Developed and optimized **Power BI** dashboards and **SQL-based reporting** solutions for business stakeholders, integrating directly with **Snowflake** for real-time data analysis.
- Used **Power BI's DirectQuery** functionality to query **Snowflake** for live analytics.
- Delivered actionable insights from **Snowflake** data warehouses, focusing on performance indicators and financial metrics for **Credit Suisse**.
- Automated data workflows and processing tasks using **Snowflake Tasks** and **Azure Data Factory** to ensure timely and consistent data loads.
- Set up performance monitoring and automated alerting in both **Snowflake** and **Azure**, utilizing **Azure Monitor** and **Snowflake's Query History**.
- Implemented cost management strategies by optimizing **Snowflake Storage** and **compute resources**, leveraging **Azure Cost Management** and **Snowflake's Resource Usage Monitoring**.

Tools Utilized: Snowflake Data Warehouse, Snowflake SQL, Snowpipe, Streams & Tasks, Virtual Warehouses, Clustering Keys, Secure Data Sharing, Resource Monitoring, Azure Data Factory (ADF), Azure Blob Storage, Azure Data Lake, Azure SQL Database, Azure Event Hubs, Azure Monitor, Power BI.

Client: TCS, India

March 2021 – Dec 2022

Role: ETL Snowflake Developer

Responsibilities:

- Designed, developed, and maintained ETL pipelines leveraging **Snowflake** for data ingestion, transformation, and storage, ensuring efficient data flow and seamless integration with external systems.
- Utilized **SQL** and **Snowflake Streams** and **Tasks** for incremental loading, data transformations, and scheduled processing, optimizing for performance and resource utilization.
- Automated and orchestrated ETL workflows using **AWS Glue** for serverless ETL tasks and **AWS Lambda** for event-driven processing, improving pipeline efficiency and reducing manual intervention.
- Architected and optimized **Snowflake** data models, including star schema and snowflake schema, for scalable, high-performance data warehousing.
- Implemented data partitioning strategies and **clustering keys** within Snowflake to enhance query performance, reducing runtime and costs associated with large datasets.
- Integrated data storage solutions with **AWS S3** for raw data storage and **Snowflake External Tables** for seamless querying of data stored in S3, ensuring efficient data access.
- Led data migration efforts, moving large volumes of data from on-premises systems (such as SQL Server and Oracle) to **Snowflake** using **AWS S3** as an intermediary storage layer and **Snowflake's Data Loading** capabilities.
- Integrated **Snowflake** with AWS-based data tools like **AWS Glue** for data transformation and **AWS Redshift** for analytical workloads, ensuring seamless interoperability between different cloud platforms.
- Implemented automation of ETL workflows using **AWS Lambda** for event-driven executions and scheduled tasks via **AWS CloudWatch Events**, reducing manual errors and improving pipeline monitoring.
- Tuned query performance by utilizing **Snowflake's Query Profile** and optimizing **SQL** queries, leveraging **Materialized Views** and **Snowflake's automatic clustering** features to improve efficiency and minimize processing time.
- Used **AWS CloudWatch Logs** to monitor, log, and troubleshoot data pipeline performance, ensuring high reliability and quick issue resolution.
- Worked closely with cross-functional teams including **Business Analysts**, **Data Engineers**, and **Data Scientists** to gather requirements and translate them into optimized ETL processes within **Snowflake** and AWS.
- Collaborated with **AWS Security** and **Cloud Operations** teams to implement secure data handling procedures, including managing **IAM Roles**, ensuring **data encryption** at rest and in transit, and defining access controls for both **Snowflake** and AWS resources.
- Designed and implemented automated validation checks within the ETL pipelines using **Python** and **SQL** to ensure data integrity and quality during data extraction, transformation, and loading.
- Created robust error handling mechanisms and alerts using **AWS CloudWatch** to monitor pipeline failures, sending notifications and implementing recovery processes to ensure data consistency.
- Integrated **Snowflake** with various **AWS services**, such as **AWS S3** for data storage, **AWS Glue** for data transformation, and **AWS IAM** for managing secure access to cloud resources, ensuring adherence to security and compliance requirements.
- Utilized **AWS CloudFormation** and **Terraform** for provisioning and managing infrastructure, streamlining the deployment and management of cloud resources.

Tools Utilized: Snowflake, SQL, AWS Glue, AWS Lambda, Snowflake Streams, Tasks, Snowflake, Clustering Keys, Partitioning, AWS S3, Snowflake External Tables, AWS Redshift, AWS CloudWatch Events, Snowflake Materialized Views, AWS IAM, Encryption, SQL, Python, AWS CloudWatch.

Client: Atria, India
2021

June 2019 – Feb

Role: Informatic Developer

Responsibilities:

- **Developed and optimized ETL workflows** using **Informatica PowerCenter**, implementing complex data transformations and ensuring high-performance data processing across large datasets for client-specific needs.
- **Designed, developed, and tested data mappings**, reusable transformations, and parameterized sessions to extract, transform, and load (ETL) data from multiple heterogeneous sources, including flat files, relational databases (**Oracle**, **SQL Server**) and web services.

- **Created and maintained complex SQL queries** for data validation, data extraction, and optimization of the ETL process, utilizing advanced SQL techniques such as joins, aggregations, and subqueries for efficient data retrieval and processing.
- Implemented **data quality checks** using built-in **Informatica Data Quality (IDQ)** features to ensure the integrity, accuracy, and consistency of data at all stages of the ETL process.
- **Developed and deployed error-handling strategies** within workflows to manage job failures, ensuring smooth execution through the use of **session logs**, email notifications, and recovery mechanisms.
- **Collaborated closely with business analysts** and stakeholders to understand requirements, map data sources, and translate business needs into scalable and maintainable ETL solutions aligned with project goals and timelines.
- **Worked extensively with relational databases** like **Oracle**, **SQL Server**, and **MySQL** to ensure seamless data extraction, loading, and transformation processes between source and target systems, utilizing tools such as **SQL Developer** and **Toad** for development and troubleshooting.
- **Created and maintained reusable transformations** and **dynamic lookups** within Informatica to increase reusability and scalability of ETL processes, ensuring consistency across different environments.
- Applied **best practices for performance tuning**, including partitioning, session optimization, and incremental data loading techniques, resulting in a reduction of job runtime and efficient resource utilization.
- **Monitored and troubleshooted ETL workflows** in production environments, responding to and resolving issues quickly through detailed analysis of session logs and logs from relational databases, minimizing downtime and ensuring timely data delivery.
- Conducted **unit testing** and **integration testing** on newly developed and modified ETL processes, ensuring that data transformations and workflows performed as expected across different datasets, achieving optimal output.
- **Documented ETL processes, workflow configurations, and database schemas** to ensure traceability, ease of future updates, and team knowledge sharing, maintaining an organized repository of design specifications and metadata.
- Worked closely with project managers to ensure **timely delivery of ETL processes**, managing priorities and adjusting timelines based on project scope changes or urgent troubleshooting.
- **Improved data extraction processes** by incorporating best practices in data parsing, cleansing, and validation, ensuring that upstream data used in ETL processes was complete, accurate, and consistent for downstream analysis.
- Utilized **Unix shell scripting** to automate repetitive tasks, schedule jobs, and manage process logs, contributing to the automation and streamlining of routine ETL operations.
- Performed **data reconciliation** and **data integrity checks** to ensure that the extracted and transformed data met business requirements, performing cross-validation with source systems to ensure alignment between source and target data.

Tools Utilized: Informatica PowerCenter, Informatica Data Quality (IDQ), SQL, PL/SQL, Oracle, SQL Server, Toad for Oracle, SQL Developer, Unix/Linux Shell Scripting, JDBC (Java Database Connectivity), Control-M.

Education:

University of North Texas

Denton, TX, USA

Master's in Computer Science and Engineering

Rajiv Gandhi University of Knowledge Technologies

Basar, TS,

INDIA

Bachelors in Electronics and Computer Science

Soft Skills:

- Problem-solving

- Team Collaboration
- Client Communication
- Adaptability
- Leadership